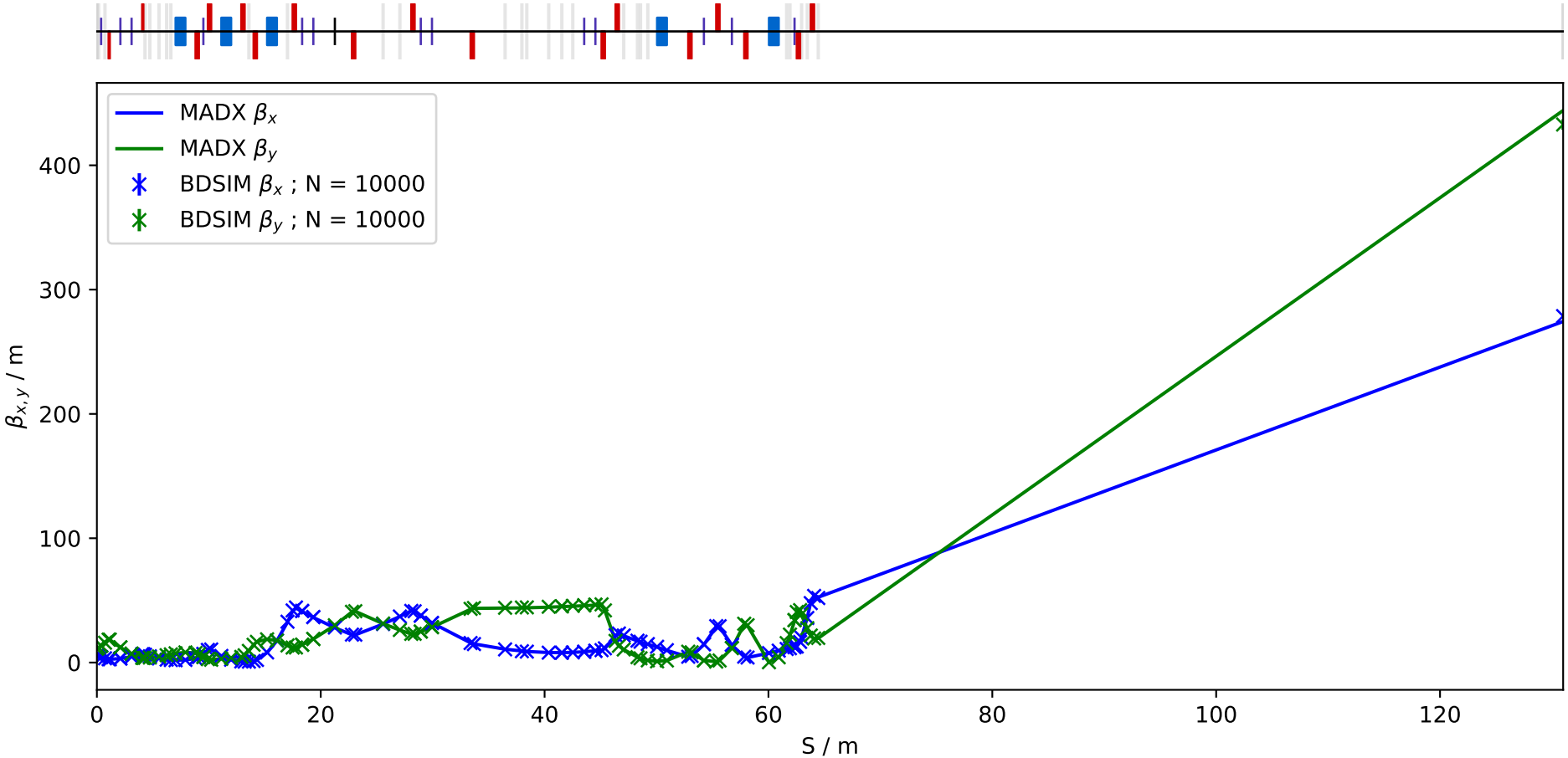
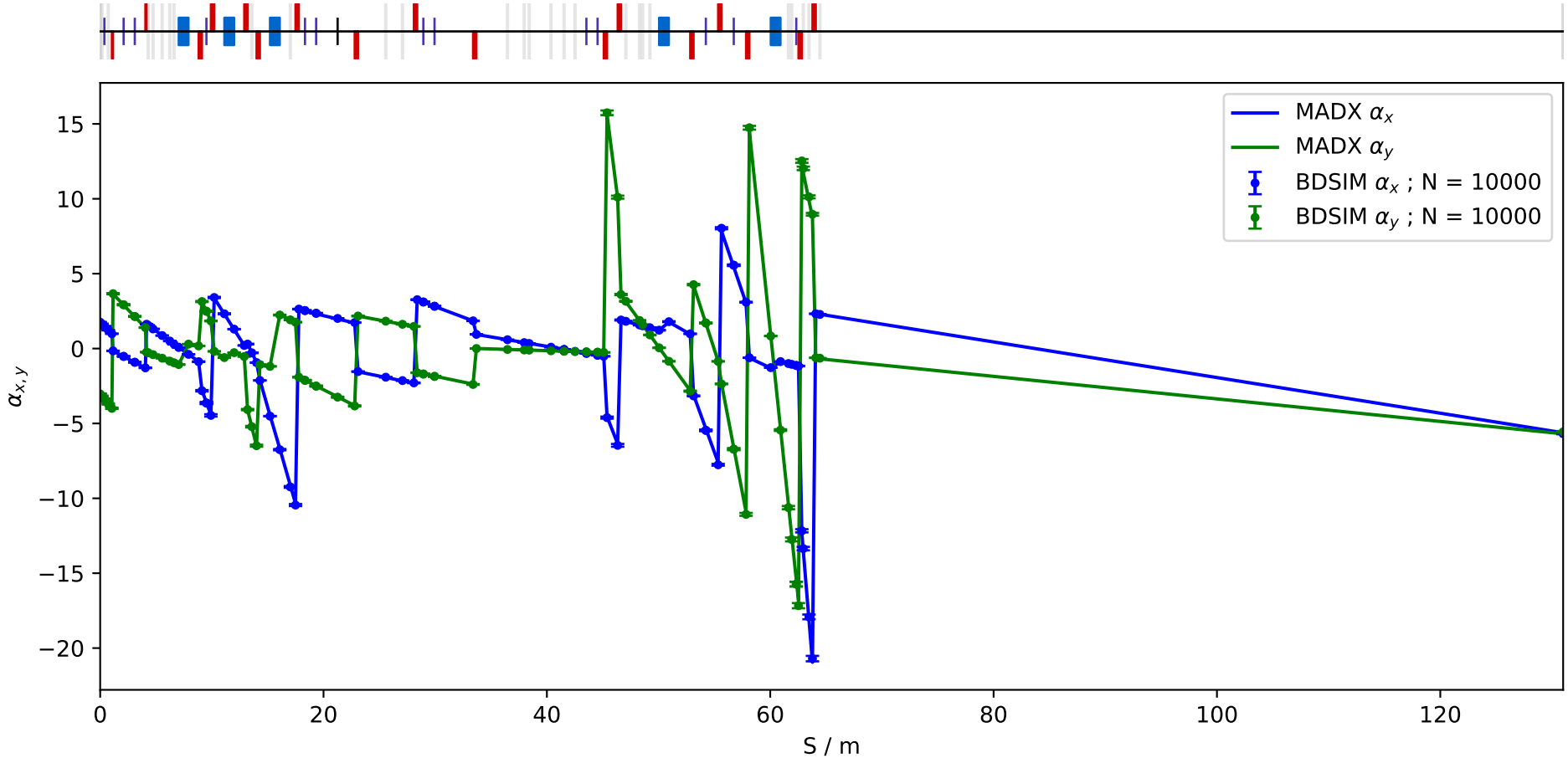
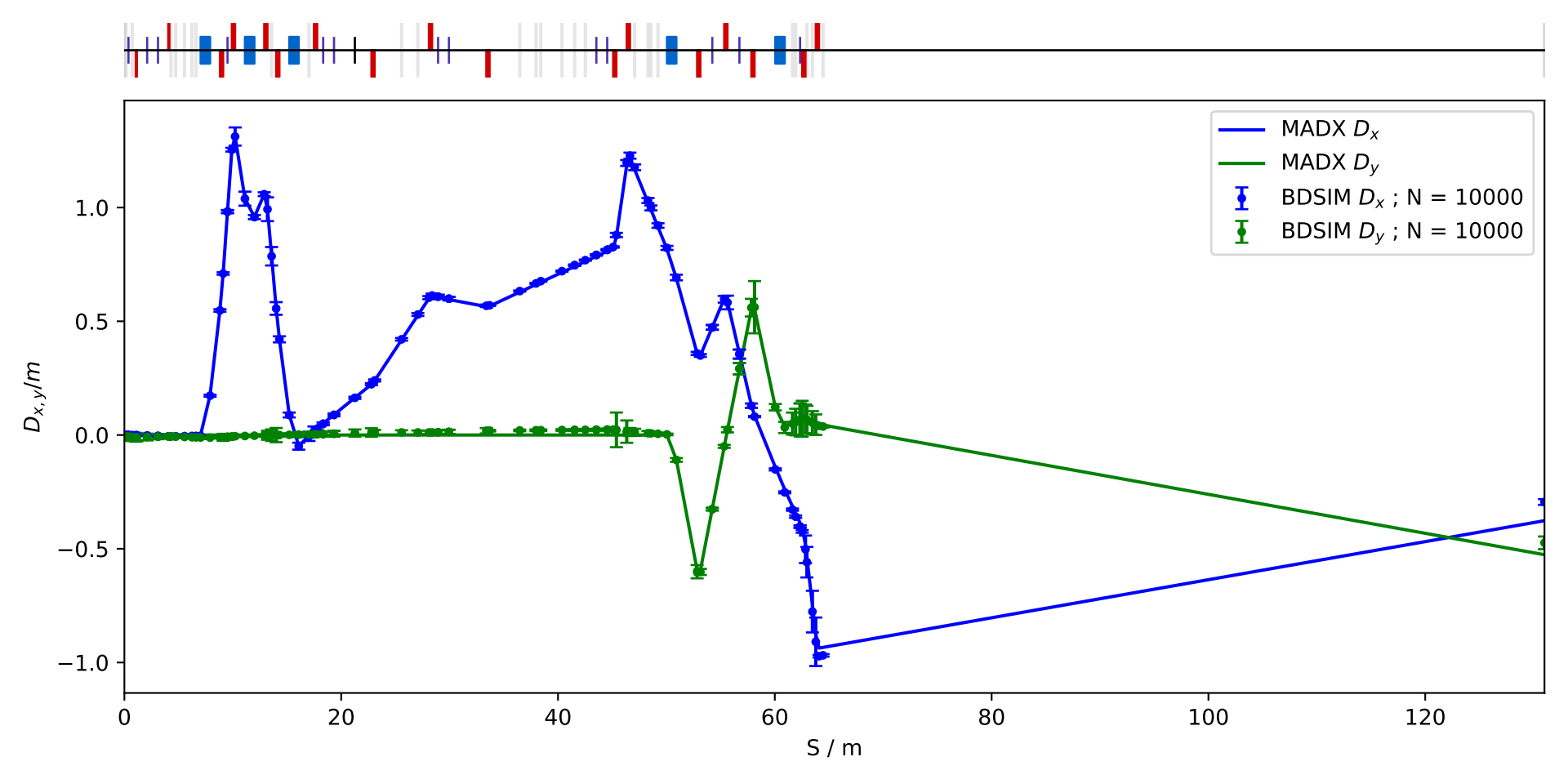
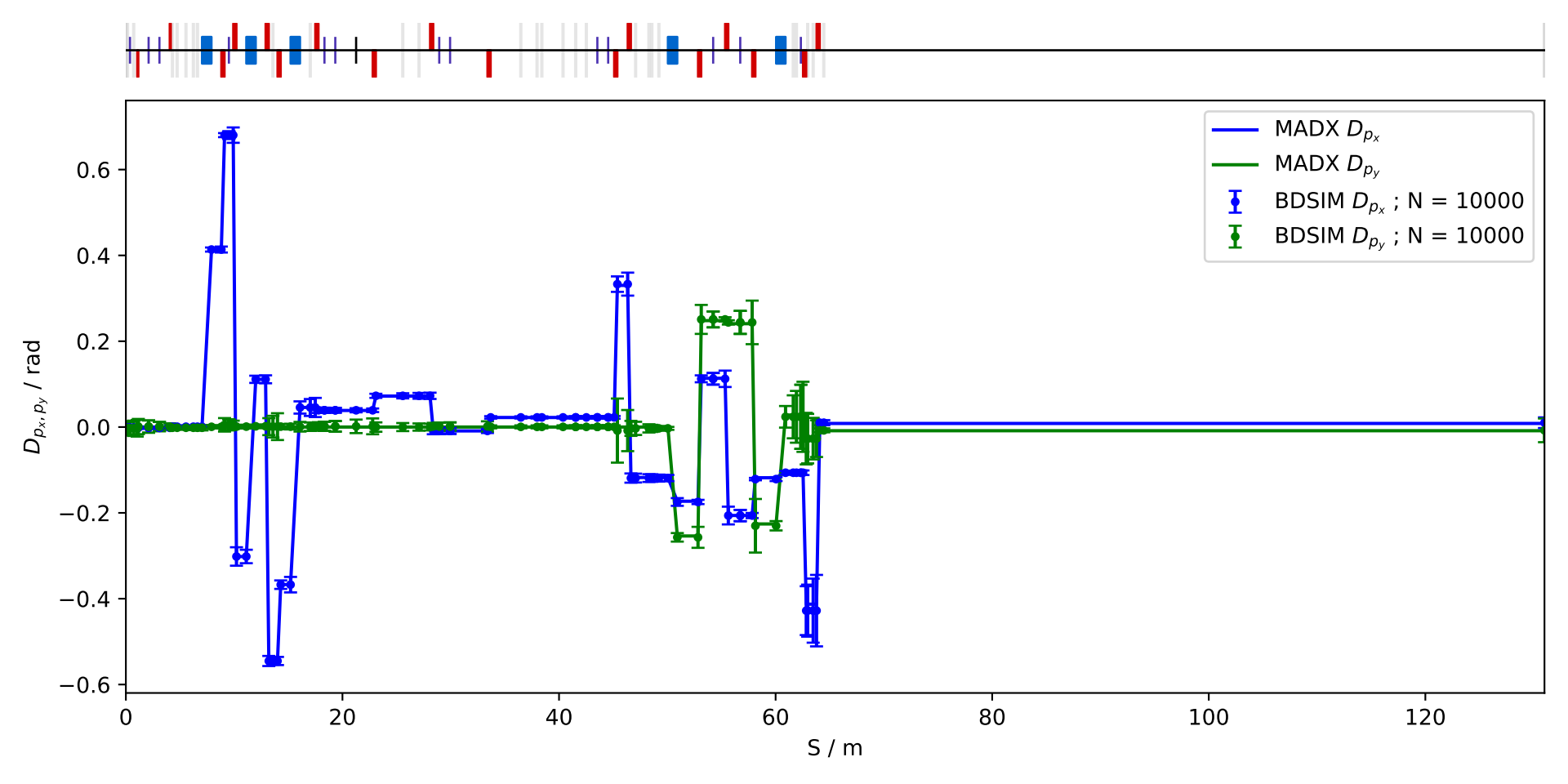
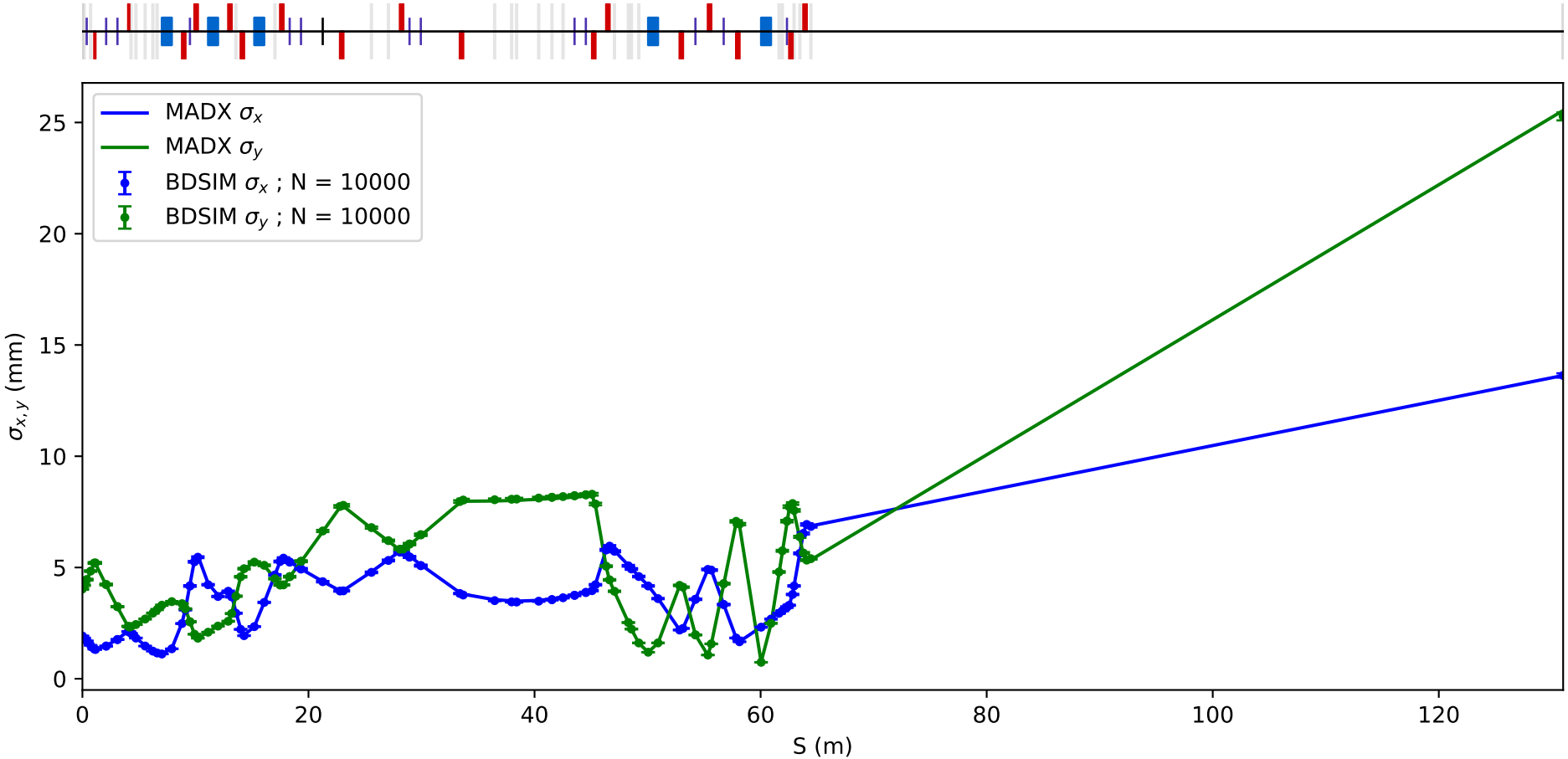
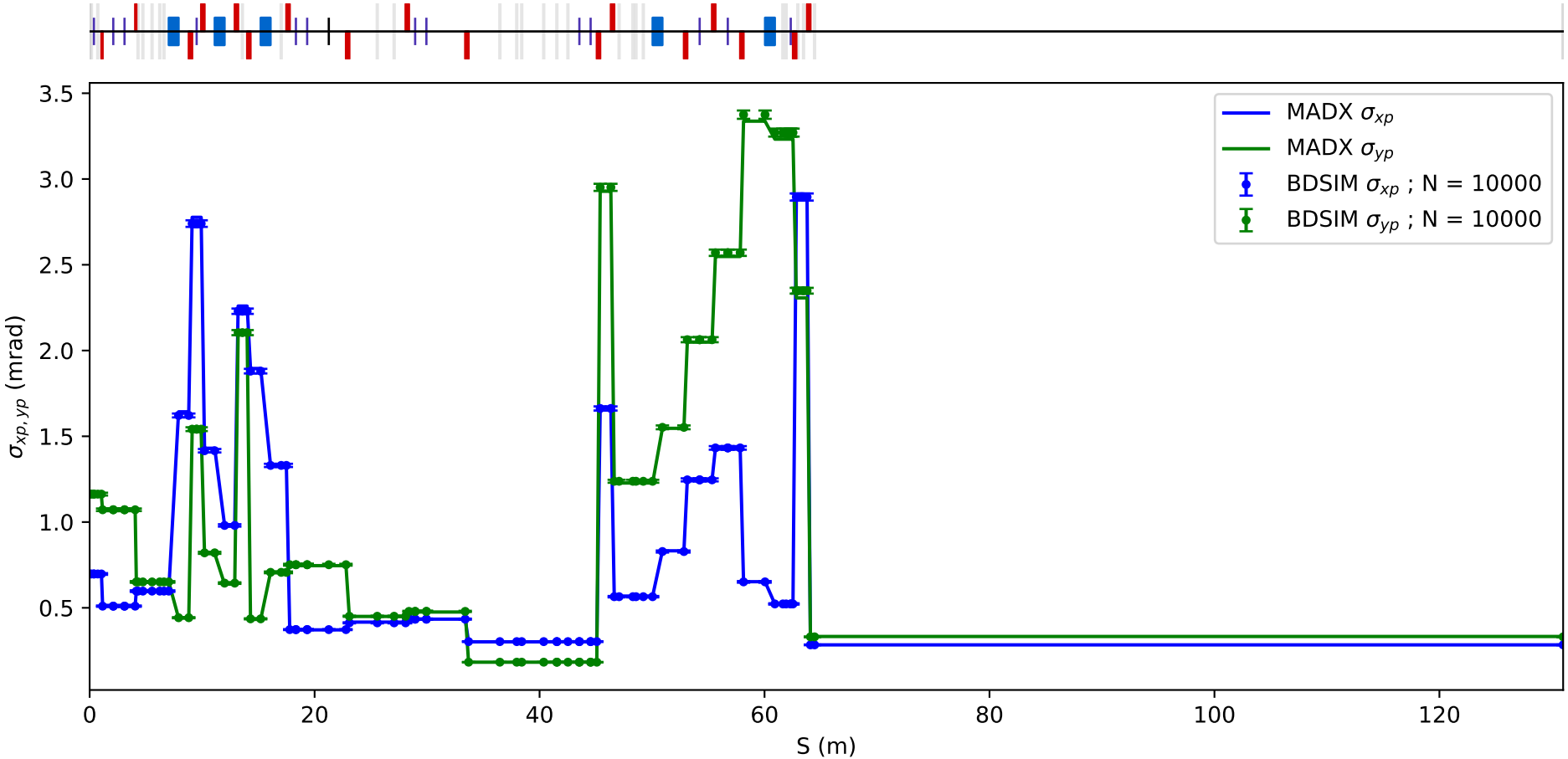

MADX β_x
MADX β_y
BDSIM β_x ; N = 10000
BDSIM β_y ; N = 10000
$\beta_{x,y}$ / m
S / m
0
100
200
300
400
0
20
40
60
80
100
120

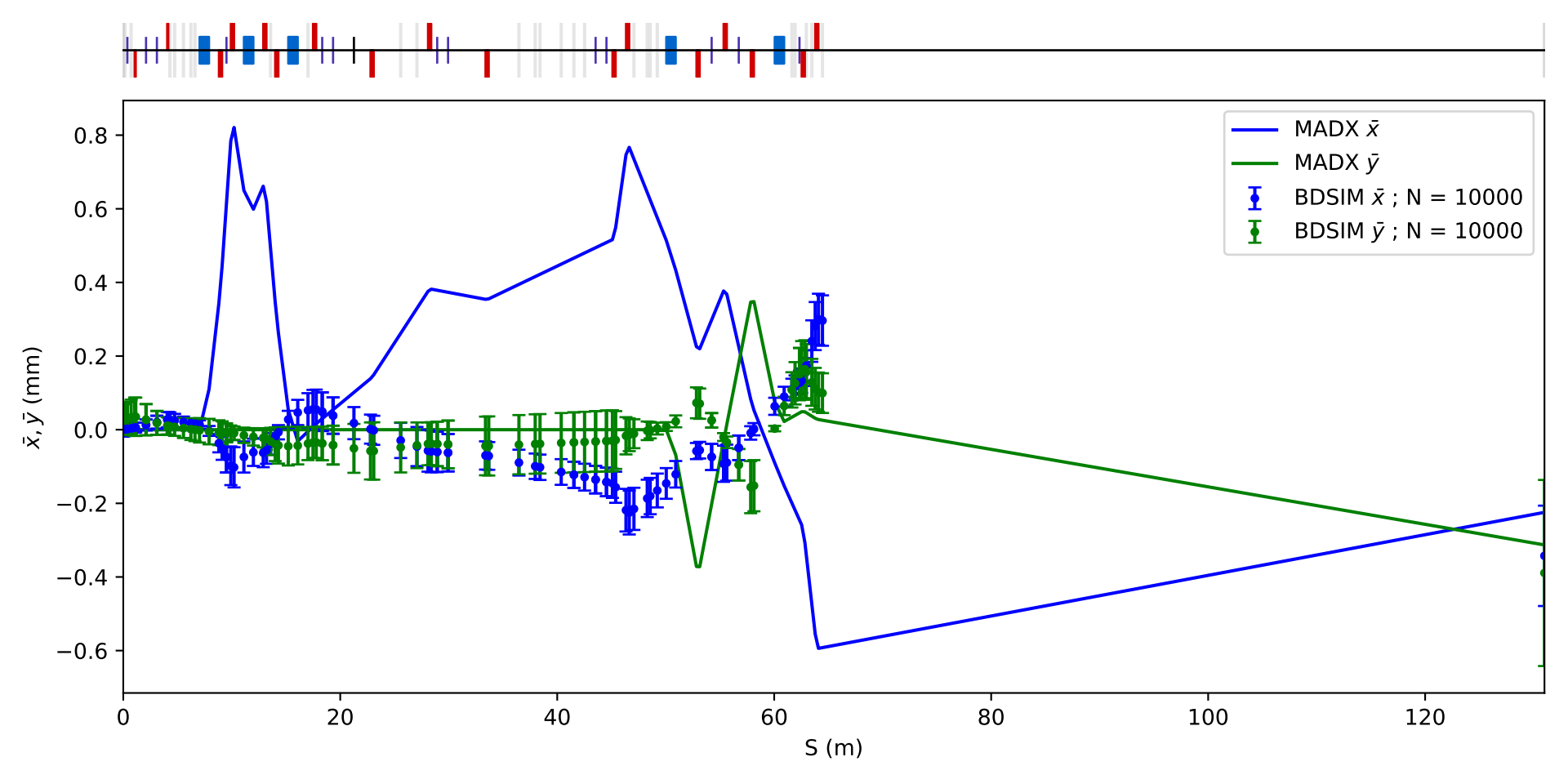

MADX α_x
MADX α_y
BDSIM α_x ; N = 10000
BDSIM α_y ; N = 10000
$\alpha_{x,y}$
S / m
15
10
5
0
−5
−10
−15
−20
0
20
40
60
80
100
120

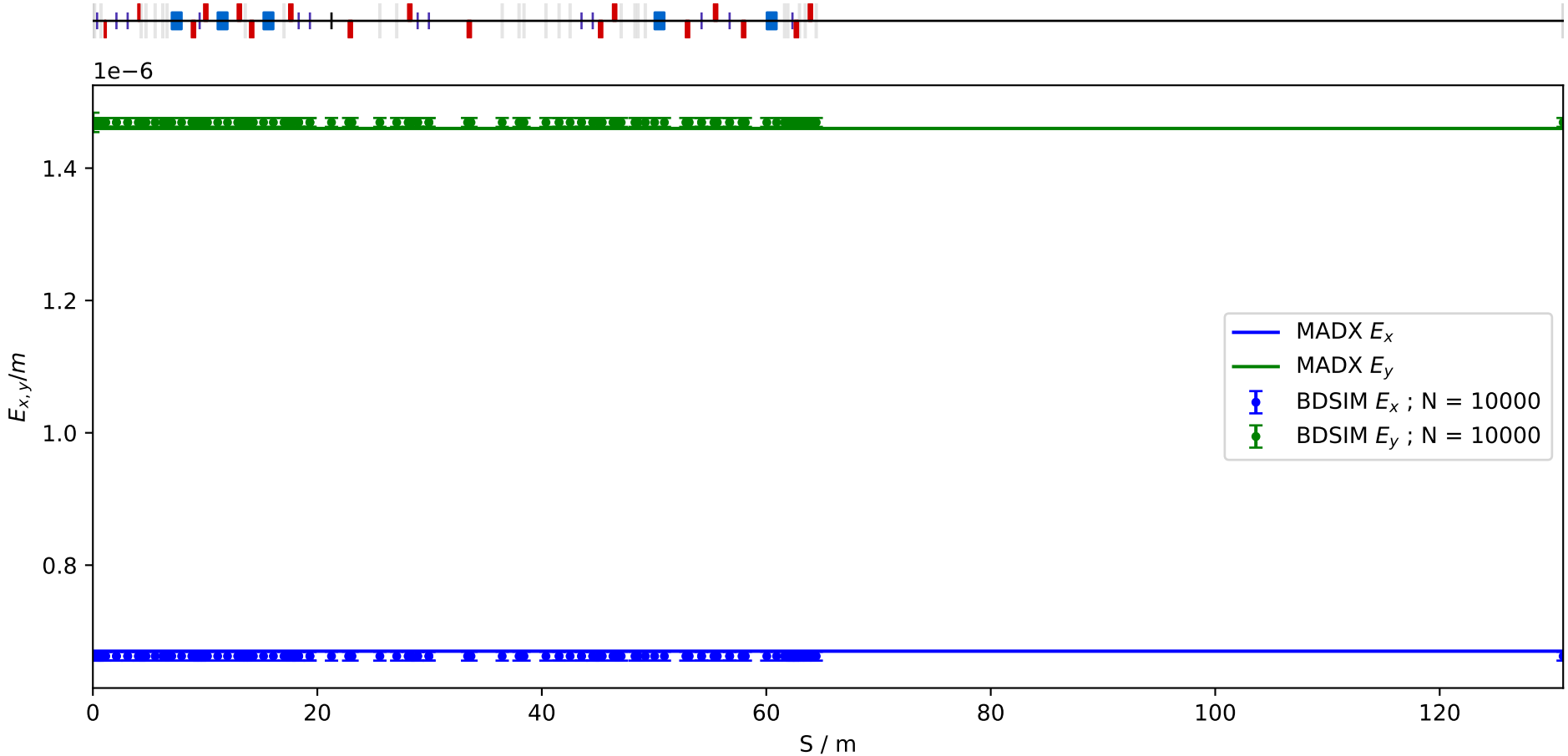

MADX D_x
MADX D_y
BDSIM D_x ; N = 10000
BDSIM D_y ; N = 10000
$D_{x,y}/m$
S / m
1.0
0.5
0.0
−0.5
−1.0
0
20
40
60
80
100
120

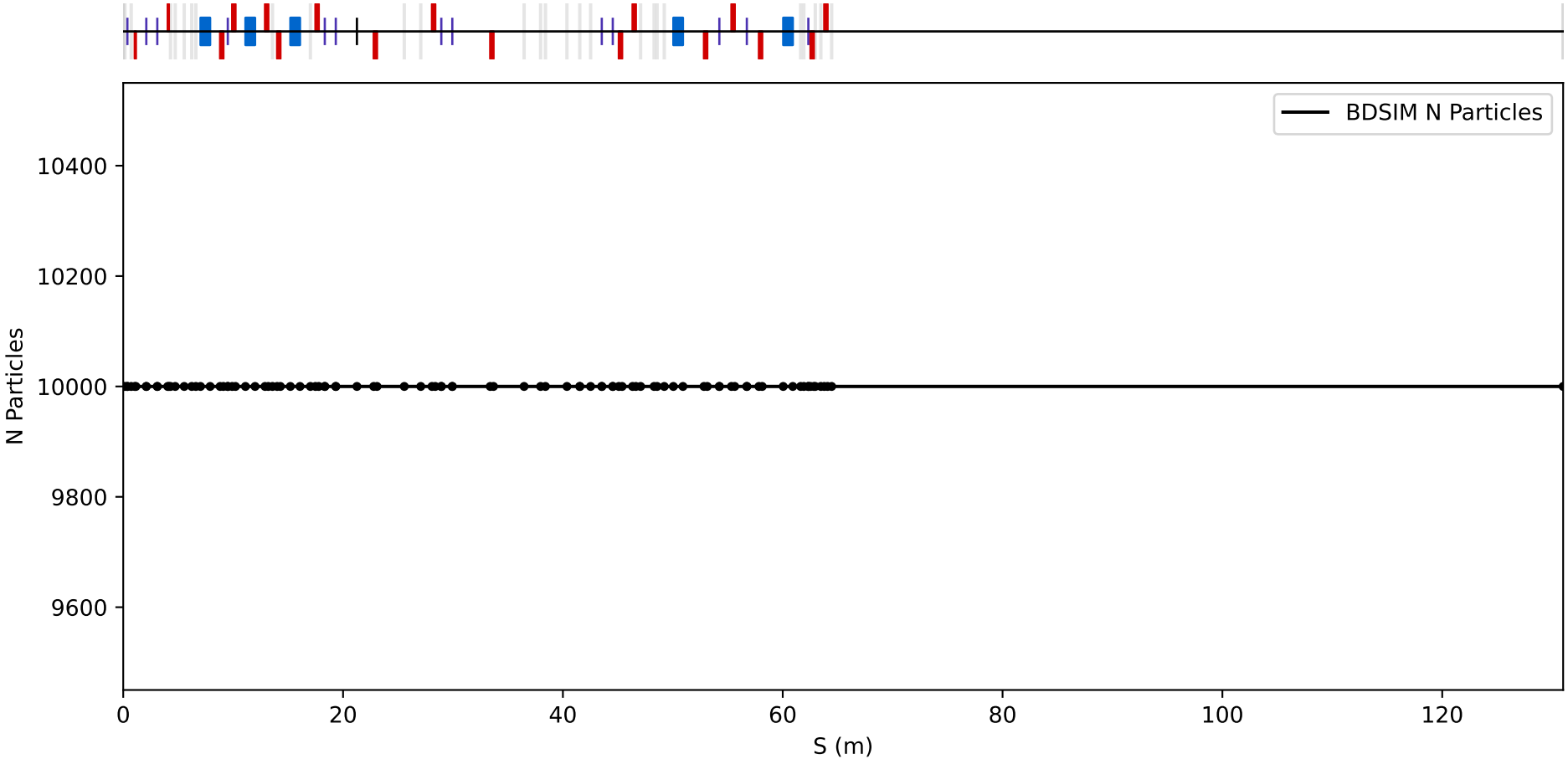

MADX D_{p_x}
MADX D_{p_y}
BDSIM D_{p_x} ; N = 10000
BDSIM D_{p_y} ; N = 10000
D_{p_x, p_y} / rad
S / m
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0
20
40
60
80
100
120


MADX σ_x
MADX σ_y
BDSIM σ_x ; N = 10000
BDSIM σ_y ; N = 10000
$\sigma_{x,y}$ (mm)
S (m)
0
5
10
15
20
25
0
20
40
60
80
100
120


MADX σ_{xp}
MADX σ_{yp}
BDSIM σ_{xp} ; N = 10000
BDSIM σ_{yp} ; N = 10000
$\sigma_{xp,yp}$ (mrad)
S (m)
0
20
40
60
80
100
120
0.5
1.0
1.5
2.0
2.5
3.0
3.5


MADX $\bar{x}$
MADX $\bar{y}$
BDSIM $\bar{x}$; N = 10000
BDSIM $\bar{y}$; N = 10000
$\bar{x}, \bar{y}$ (mm)
S (m)
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0
20
40
60
80
100
120


1e−6
1.4
1.2
1.0
0.8
$E_{x,y}/m$
0
20
40
60
80
100
120
S / m
MADX E_x
MADX E_y
BDSIM E_x ; N = 10000
BDSIM E_y ; N = 10000


BDSIM N Particles
10400
10200
10000
9800
9600
N Particles
0
20
40
60
80
100
120
S (m)